
SEIF EDDINE JOUL

Mechatronics Engineer

Morocco

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SUMMARY

Multidisciplinary engineer proficient in instrumentation, PLC logic development, and diagnostic troubleshooting, strengthened via hands-on lab experimentation and advanced projects. Highly Skilled in integrated systems design as well as core electrical & mechanical systems commissioning. I am enamored with innovation, to me engineering is all about novel and efficient problem-solving and optimizing existing solutions. Passionate about getting my hands dirty to turn theory and ideas into working reality; aiming for the beauty of robust, automated, and maintainable system design, your teams will definitely feel my impact. I am flexible, ready to relocate, and thrive in both fieldwork and office-based environments. My reputation? Active team-player and reliable solo achiever.

EXPERIENCE

Freelancer - *Independent, Morocco*

Apr 2025 - PRESENT

- Electrician in a project-based environment, working on-site renovation and construction projects, specializing in wiring, lighting, and power distribution; and ensuring documentation and safety standards are met.
- Specialized in installation of ELV systems (CCTV surveillance), handling power supply and data transfer and network integration, in commercial and residential settings. With client onboarding and technical handover to ensure operational readiness.

Ast. Electromechanical engineer - *Hitronik, Malaysia*

Aug 2023 - Oct 2025

- Personally handled restoration of an old CNC lathe machine, as part of a team handling maintenance and commissioning duties for university equipment and lab machinery.
- Co-led the development and commissioning of multiple solar powered automated systems using repurposed mechanical and electrical components, ensuring compatibility between parts and promoting sustainability.
- Managed electrical circuit schematic design and assembly from scratch, performing component level soldering and mechanical coupling, along with drafting rough ICDs for each project.

Deputy President- *USM International Student Society, Malaysia*

Feb 2023 – Apr 2025

- Introduced a collaborative & results-driven approach during my tenure, creating new partnerships with adjacent campus societies and departments, significantly boosting our presence & reputation.
- Worked directly with our director to lead my team to plan and execute several large-scale multicultural initiatives and events that consistently advancing the society's vision and purpose, successfully achieving our annual objectives each time.
- Significantly increased our organizations outreach and visibility via increased social media presence, targeted marketing methods and focusing on a more personal approach regarding our initiatives; confirmed a x3 increase in social media monthly engagement and event turnout.

EDUCATION

Bachelor in Mechatronics Engineering - *Universiti Sains Malaysia, Malaysia* Feb 2020 - Nov 2024

Grad with an Honours from a university ranked #134 globally (QS). Outcome-Based Education program accredited under the international Washington Accord. Specialized in automation and instrumentation. Electives in Computer Vision and Quality, active athlete and volunteer on campus.

High school Diploma - *Al Nahda National School, UAE*

Jan 2016 - Jun 2019

SKILLS & CERTIFICATES

Skills: SolidWorks, AutoCAD, MATLAB, Microcontrollers, Instrumentation, Microsoft Office Suite, Problem Solving, LabVIEW, KiCAD, PCB Design, OpenCV, Python, C++, Schematic Design, ELV systems, IEC 61131-3 languages, Power Distribution, Systems Integration, Field Diagnostics & Troubleshooting

Certificates in: Manufacturing engineering drawing, Sustainability Manufacturing, Technical writing, Communication protocols, IELTS (C1/7), Computer Vision

PROJECTS

LiDAR-Camera Sensor Fusion System – *Computer Vision/Sensor Fusion*

Developed a MATLAB-based LiDAR-camera fusion system using extrinsic calibration, projective transformation, and YOLO object detection. Achieved sub-pixel alignment (<0.03 px reprojection error) and improved obstacle detection and depth estimation accuracy by $\sim 35\%$.

4-DOF Robotic Arm (Simulation /Fabrication /Control) – *Robotics /Control Systems*

Designed and fabricated a servo-actuated robotic arm integrating forward/inverse kinematics, Cartesian motion control, and custom GUI interface using Arduino, LabVIEW, and SolidWorks. Achieved end-effector positioning accuracy within ± 3 mm and response latency below 100 ms.

Regenerative Braking System for Amphibious Robot – *Power Electronics*

Designed a regenerative braking mechanism integrating LiPo recharge logic, RPM-based energy capture calibration, and sensor-triggered braking control. Reduced simulated energy losses by up to 25% across operating speeds from 30–1530 rpm.

Vacuum Pick-and-Place Handling Station – *Industrial Automation /Pneumatics*

Designed and commissioned a PLC-controlled pneumatic pick-and-place station using ladder logic AND structured text on OMRON CX-ONE. Achieved repeatable transfer cycles under 5 seconds with 0.5 mm positioning accuracy while improving system reliability by $\sim 70\%$.

Automated Solar Irrigation & Smart Signage System – *Renewable Energy /Sustainable Systems*

Designed and commissioned an autonomous solar-powered irrigation and LED signage system. Handled PV installation, custom PCB circuitry, mechanical coupling, and power storage; using repurposed components modified for compatibility, improving water-flow efficiency and energy autonomy by $\sim 60\%$.

International Food and Culture Fiesta – *Leadership/Event Management*

Led the organization of a multicultural university event hosting 500+ attendees with a managed budget exceeding RM 20,000. Coordinated logistics, media, sponsorships, and team operations to deliver one of the university's flagship student events.

Drone Delivery Optimization & Reliability Modeling – *Data Analysis /Reliability Engineering*

Conducted drone delivery simulations using Minitab and one-way ANOVA across 5 payload platform configurations with 30+ simulation runs each. Applied regression modeling, stress analysis, and Weibull prediction techniques to evaluate multiple metrics, identifying a 2x increase in delivery time from platform size 2 to 10 while achieving $>90\%$ statistical confidence in performance behavior predictions.